

# tf.compat.v1.nn.ctc\_beam\_search\_decoder

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*Performs beam search decoding on the logits given in input.*

## Function Signatures

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```
tf.compat.v1.nn.ctc_beam_search_decoder( inputs,  
sequence_length, beam_width=100, top_paths=1,  
merge_repeated=True )
```

## Code Examples

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```
tf.compat.v1.nn.ctc_beam_search_decoder(  
inputs, sequence_length, beam_width=100, top_paths=1,  
merge_repeated=True  
)
```

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inputs, sequence_length, beam_width=100, top_paths=1,  
merge_repeated=True  
)
```

# Documentation

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Performs beam search decoding on the logits given in input.

- `ctc_beam_search_decoder` treats blanks as sequence termination
- `ctc_greedy_decoder` treats blanks as regular elements

If `merge_repeated` is `True`, merge repeated classes in the output beams.

This means that if consecutive entries in a beam are the same, only the first of these is emitted. That is, when the sequence is `A B B * B * B` (where `'*'` is the blank label), the return value is:

- `A B` if `merge_repeated = True`.
- `A B B B` if `merge_repeated = False`.

## Returns

`decoded[j].indices`: Indices matrix (`total_decoded_outputs[j]` x 2)

The rows store: `[batch, time]`.

`decoded[j].values`: Values vector, size (`total_decoded_outputs[j]`).

The vector stores the decoded classes for beam `j`.

`decoded[j].dense_shape`: Shape vector, size (2).

The shape values are: `[batch_size, max_decoded_length[j]]`.